

Introduction: Application of Applied Behavior Analysis (Internal therapist Training)

Autism Spectrum Disorder (ASD)

Autism Spectrum Disorder (ASD) is defined by the Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition (DSM-5). Diagnosis is based on two primary domains:

- A. Social communication and social interaction deficits
 - a. Deficits in social-emotional reciprocity
 - b. Deficits in nonvocal communicative behaviors used for social interaction
 - c. Deficits in developing, maintaining, and understanding relationships
- B. Restrictive, repetitive patterns of behavior, interest, or activities
 - a. Stereotyped or repetitive motor movements, use of objects, or speech
 - b. Insistence on sameness, inflexible adherence to routines, or ritualized patterns of verbal or nonverbal behavior
 - c. Highly restricted, fixated interests that are abnormal in intensity or focus
 - d. Hyper- or hyporeactivity to sensory input or unusual interest in the sensory aspect of the environment

★ **Note:** Symptoms must present in the early development period and cause significant impairment in everyday functioning. Symptoms are not better described by another DSM5 diagnosis.

Common Characteristics & Deficits

- Communication delays (verbal and nonverbal)
 - Early Sign: Lack of gestures (e.g., pointing, waving), Delayed speech or language regression
- Difficulty with social interactions
 - Early Sign: Limited eye contact or response to name, Reduced interest in social games or peers
- Restricted interests and repetitive behaviors
 - Early Sign: Repetitive movements or play
- Sensory sensitivities (e.g., noise, textures)
- ★ Range of abilities and challenges (no “one size fits all”)

Note:

Ethical and Professional

Code of Ethics

Scope of Competence:

- Practice Limits: Practice only within the scope of competence as established by your education, training, and experience.
- Supervision: All services must be provided under appropriate professional supervision.

Integrity:

- Maintain the highest standards of professional behavior and act in the best interest of the client at all times.
- Demonstrate trustworthiness, honesty, fairness, and sincerity in all professional interactions.

Confidentiality:

- Privacy: All information regarding clients—including identifying information, diagnosis, assessment, treatment, and prognosis is strictly confidential.
- Compliance: Staff must comply with all privacy rules of the country and the company's internal policies.
- Digital Privacy: Sharing social media content containing client-related information, disclosing client records, data, photographs, or videos, and discussing clients with unauthorized individuals are strictly prohibited.

Dignity and Respect:

- Behave toward others with compassion, dignity, and respect.
- Equitable Treatment: Treat all individuals fairly regardless of age, disability, ethnicity, gender identity, immigration status, marital status, national origin, race, religion, sexual orientation, or socioeconomic status.
- Self-Determination: Respect and actively promote the client's right to self-determination, especially when working with vulnerable populations.
- Informed Choice: Acknowledge that personal choice is vital; provide clients and stakeholders with the necessary information to make informed choices about their services.

Disclosure:

- Disclosure of confidential communication is prohibited unless authorized in writing or mandated by law to avoid a serious threat to the health and safety of the client or others.

Informed Consent:

- Informed consent is the process of obtaining a written agreement from stakeholders that demonstrates their ability and willingness to participate in related services or guidelines.

Work Product and Records:

- All client-related work products remain confidential and must be securely archived for up to 7 years after the conclusion of treatment.

Protection from Harm:

- Safety First: Protect the client from harm throughout the entire duration of treatment.
- Safe Reinforcers: Choose reinforcers that are not potentially hazardous to the client's health or safety.

- Individualization: Always take the client's specific physical and cognitive abilities into consideration when selecting reinforcers.
- Child Abuse & Mandated Reporting: Staff are legally and ethically required to recognize signs of physical abuse, neglect, and human trafficking.
 - Any suspicion of misconduct or maltreatment must be reported immediately to a supervisor.
 - *For a comprehensive guide on identifying specific signs, symptoms, and the full reporting protocol, please refer to "Appendix 1: Child Abuse & Mandated Reporting" at the end of this manual.*

Client Rights:

- Be aware of the client's legal and ethical rights; take immediate action if those rights are violated or at risk of violation.
- Transparency: Clients have the right to receive accurate data, reports, and progress information that is jargon-free and presented in a clear, comprehensible manner.

Mandated Reporting:

- Suspected misconduct or known abuse/neglect of a child, elder, or dependent adult must be immediately reported to a supervisor.

Human Relations:

- **Avoiding Harm:** Take all reasonable steps to avoid or minimize harm to clients, supervisees, and colleagues.
- **Unfair Discrimination:** Do not engage in discrimination based on any protected status or personal characteristic.
- **Harassment:** Do not engage in demeaning or harassing behavior, including any form of sexual harassment, toward subordinates, clients, or colleagues.
- **Multiple Relationships and Conflicts of Interest**
- **Boundaries:** Avoid multiple relationships (dual roles) with clients, supervisors, or stakeholders that could impair objectivity.
- **Intimacy:** Avoid sexual or intimate relationships with clients, their family members, or supervisees.
- **Former Clients:** Relationships with former adult clients should be avoided for a minimum of two years.
- **Gift Policy:** To maintain professional boundaries, avoid accepting gifts valued over 500 Baht (per CARE policy).
- **Conflict of Interest:** Refrain from professional roles where personal, financial, or legal interests could impact your effectiveness or competence.

Note:

Core Principle of ABA

WHAT IS ABA?

- ABA stands for Applied Behavior Analysis
- **Definition:** Application of the principles of behavior to issues that are socially important in an effort to produce practical change
- **Core Principle:**
 - The consequences that follow a behavior control whether that behavior will increase or decrease
 - The ABCs: The 3-Term Contingency:
 - A. Antecedent
 - B. Behavior
 - C. Consequence
- ★ **Applying ABA to the treatment of Autism:**
 - We change antecedents and consequences to increase **Appropriate Behaviors & Skills**. This is the **Skills Repertoire Building** component of our program. There are 8 curricula at CARE that we are focusing on:
 - Language
 - Play
 - Social Skills
 - Perspective Taking
 - Executive Functioning
 - Motor Skills
 - Adaptive Skills
 - School Skills
 - We change antecedents and consequences to decrease **Inappropriate Behaviors**. This is the **Behavior Management** component of our program. For example
 - Stereotypy
 - Non-compliance
 - Tantrums
 - Aggression
 - Self-Injury
- ★ For a deeper look into the scientific foundations and historical characteristics of our field, please see **Appendix 2: Branches of Behavior Analysis**.

Note:

Foundational Behavioral Terminology

Extinction: The discontinuing of a reinforcement of a previously reinforced behavior

Extinction Burst: An increase in the frequency or intensity of responding when an extinction procedure is initially implemented

Spontaneous Recovery: A behavioral effect associated with extinction in which the behavior suddenly begins to occur after its frequency has decreased to its prereinforcement level or stopped entirely.

Deprivation: The state of an organism with respect to how much time has elapsed since it has consumed or contacted a reinforcer

Satiation: A decrease in the frequency of operant behavior presumed to be the result of continued contact with or consumption of a reinforcer that has followed the behavior; also refers to a procedure for reducing the effectiveness of a reinforcer

Stimulus: An environmental change that affects an organism through sensory perception.

Discriminative Stimulus (SD): A stimulus in the presence of which a given behavior has been reinforced and in the absence of which that behavior has not been reinforced; as a result of this history, an SD evokes operant behavior because its presence *signals the availability of reinforcement*

Stimulus Control: A situation in which the frequency, latency, duration, or amplitude of a behavior is altered by the presence or absence of an antecedent stimulus

Responses: A single instance or occurrence of a specific class or type of behavior

Motivation Operations (MO): An environmental variable that (a) alters (increases or decreases) the reinforcing or punishing effectiveness of some stimulus, object, or event; and (b) alters (increases or decreases) the current frequency of all behavior that has been reinforced or punished by that stimulus, object, or event.

Establishing Operations (EO): A motivating operation that increases the effectiveness of some stimulus, object, or event as a reinforcer.

Settings Events: Setting events are background factors that influence behavior. They don't trigger behavior directly but make certain behaviors more likely. It can be environmental, biological, or social

Note:

Reinforcement VS Punishment

Reinforcement: A basic principle of behavior describing a response – consequence functional relation in which a response is followed immediately by a stimulus change that results in similar responses occurring more often

- **Contingent Reinforcement:** Describes reinforcement (or punishment) that is delivered only after the target behavior has occurred
- **Non-Contingent Reinforcement:** Describes reinforcement (or punishment) that is delivered regardless of behavior

Punishment: A basic principle of behavior describing a response – consequence functional relation in which a response is followed immediately by a stimulus change that decreases future occurrences of that type of behavior.

Schedules of Reinforcement

Definition: A rule specifying the environmental arrangements and response requirements for reinforcement; a description of a contingency of reinforcement.

Types of Reinforcement Schedules:

- **Continuous Reinforcement (CRF):**
 - Provides reinforcement for each occurrence of behavior.
 - Example: A teacher using a continuous schedule of reinforcement would praise every time a student answers questions correctly
- **Intermittent schedules of reinforcement (INT):**
 - Some, but not all, occurrences of the behavior are reinforced.
 - There are four basic types of intermittent schedules of reinforcement
 1. **Fixed Ratio (FR):** Reinforce after a set number of responses
 - a. Example: FR5: Reinforce every 5 responses
 2. **Variable Ratio (VR):** Reinforce after an average number of responses
 - a. Example: FV4: Reinforce about every 3-5 times (average of 4 times)
 3. **Fixed Interval (FI):** Reinforce after a set amount of time
 - a. Example: FI2: Reinforce the first response after 2 minutes
 4. **Variable Interval (VI):** Reinforce after a varying time
 - a. Example: VI3: Reinforce the first response after 2–4 mins (average of 3 minutes)

Note:

Skill Acquisition

Discrimination Training

Definition: The procedure for teaching the child to distinguish between 2 or more SDs

Terms in Discrimination Training:

- **Acquisition Item/Target:** Term used to describe the item(s) that you are currently teaching
- **Field:** Array of items/stimuli for the client to pick, point, or touch as a correct response
- **No Field:** No stimuli/item for the client, the correct response needs to be performed or vocally response, such as clap, or answer a question, "I'm 5 years old."
- **Distracter item:** Additional items that are not targets, used to teach simultaneous discriminations
- **Distracter trial (DT)-** Additional trials that do not target the acquisition item are used to teach successive discriminations
- **Mass Trial (MT):** Repeated presentation of one SD
- **Expanded Trials (ET):** Systematically increasing the time between presentations of the target SD
 - Order of SDs is not random; there is a pattern
 - Can use any and many previously mastered items during distracter trials
 - Example:
 - ET1: Target X, **D**, Target X, **D**, Target X, **D**, Target X
 - ET2: Target X, **D**, **D**, Target X, **D**, **D**, Target X, **D**, **D**, Target X
 - ET3: Target X, **D**, **D**, **D**, Target X, **D**, **D**, **D**, Target X, **D**, **D**, **D**, Target X
- **Random Rotation (RR):** Presentation of 2 or more SDs in random order
- **Graduated Random Rotation (GRR):** Systematically introducing mastered items (one at a time) into random
 - Order of SDs is random
 - Can only introduce one additional previously mastered item in each phase of GRR
 - Example:
 - RR1: RR Target X with 1 previously mastered item
 - RR2: RR Target X with 2 previously mastered items
 - RR3: RR Target X with 3 previously mastered items
- **Mastery:**
 - When the target response is 80%-100% correct, unprompted in random rotation
 - Across two or more consecutive therapy sessions
 - Across at least two therapists
 - First trial is UNPROMPTED

Note:

Types of Discrimination Training (2 Types):

1. **Simultaneous Discrimination Training (w/Field):** When the child must respond to a field of stimuli by pointing to, touching, or picking up an item in the field
 - a. When teaching the first two targets (7 steps)
 - i. **Step 1:** Mass Trial (MT) target 1
 - ii. **Step 2:** Mass Trial (MT) target 1 with 1-2 unknown distracter items
 - iii. **Step 3:** Mass Trial (MT) target 2 alone
 - iv. **Step 4:** Mass Trial (MT) target 2 with 1-2 unknown distracter items
 - v. **Step 5:** Mass Trial (MT) target 1 with target 2 as the distracter item
 - vi. **Step 6:** Mass Trial (MT) target 2 with target 1 as the distracter item
 - vii. **Step 7:** Randomly Rotate (RR) target 1 & target 2
 - b. When teaching the 3rd target and so on (4 steps)
 - i. **Step 1:** MT target “X” alone
 - ii. **Step 2:** MT target “X” with 1-2 unknown distracter items
 - iii. **Step 3:** MT target “X” with 1 or more mastered distracter items
 - iv. **Step 4:** RR target “X” with 2 or more mastered items across 2 or more consecutive sessions, with target at 80% - 100% correct unprompted
2. **Successive Discrimination Training (w/o Field):** When the child is not required to point to, touch, or pick up something from a field of stimuli
 - a. When teaching the first two targets (3 steps)
 - i. **Step 1:** Mass Trial (MT) target 1 alone
 - ii. **Step 2:** Mass Trial (MT) target 2 alone
 - iii. **Step 3:** Randomly rotate (RR) targets 1 and 2 until 80%-100% correct unprompted across 2 or more consecutive sessions
 - b. When teaching the 3rd target and so on (2 steps)
 - i. **Step 1:** MT target “X” alone
 - ii. **Step 2:** RR target “X” with 2 or more mastered items across 2 or more consecutive sessions, with target at 80% - 100% correct unprompted

Note:

Prompting

Definition:

- **Prompting:** A stimulus presented in addition to the SD that facilitates a correct response (a hint)
- **Fading:** the systematic removal of a prompt across successive trials in an effort to transfer stimulus control from the prompt to the SD

Types of Prompts:

1. **Physical:** The therapist provides physical guidance to ensure the learner completes the task.
 - a. SD: "Arms up"
 - i. Trial 1: raise child's arms over head – full physical prompt
 - ii. Trial 2: Raise arms slightly & child completes response – partial physical prompt
 - iii. Trial 3: Tap child's arms & child completes response – partial physical prompt
 - iv. Trial 4: Child responds independently
2. **Gestural:** The therapist uses a physical movement—like pointing, reaching, or nodding—to direct the learner to the correct response.
 - a. SD: "Put behind"
 - i. Trial 1: The therapist points behind the chair
 - ii. Trial 2: The child responds correctly
3. **Model:** The therapist demonstrates the target behavior for the learner to imitate.
 - a. SD: "Touch your back."
 - i. Trial 1: Therapist models the response by touching her back
 - ii. Trial 2: Therapist lightly touches her own back
 - iii. Trial 3: Child responds correctly
4. **Vocal Prompt:** Echoic - The therapist says the entire correct answer out loud.
 - a. SD: Present preferred item
 - i. Trial 1: Therapist says "I want blue" - full echoic prompt
 - ii. Trial 2 & 3: Therapist says "I..." - partial echoic prompt
 - iii. Trial 4: Jimmy responds independently
5. **Textual:** a written hint, such as a word, a list, or a script, that helps a learner perform a task or say a specific word.
 - a. The therapist holds up an unknown item
 - i. Trial 1: Therapist shows a child a card with the response, "What is it?" written on it
 - ii. Trial 5: Therapist shows the child a card with the response, "What is ___?" written on it
 - iii. Trial 8: Therapist shows the child a card with the response, "What ___ ___?" written on it
 - iv. Trial 10: Therapist shows child a card with, "___ ___ ___" written on it
 - v. Trial 12: Child responds independently
6. **Visual Prompt:** The therapist uses a picture, icon, or written word to help the learner understand what to do.
 - a. Example: Showing a picture card of a spoon while giving the instruction.
7. **Vocal Prompt: Directive** - a vocal hint that tells the learner what to do or how to do it after the initial instruction has been given.
 - a. SD: A ball is rolled to the child
 - i. Trial 1: The therapist says, "Push" – full directive prompt
 - ii. Trial 2: The therapist says "P.." – partial directive prompt
 - iii. Trial 3: The child responds independently

★ More prompts that can be used

8. **Proximity:** The therapist places the correct item closer to the learner to make the correct choice stand out.
 - a. SD: “Give me the one you measure things with.”
 - i. Trial 1: Pushes the tape measure toward the child
9. **Receptive:** a non-verbal hint that guides the learner to perform a specific movement or select the correct item. It focuses on the learner's ability to understand and "receive" the instruction by physically responding.
 - a. SD: Show item & ask, “What is it?”
 - i. Trial 1: Therapist presents receptive trial (“Give me a spatula”) immediately followed by an expressive trial (“What is it?”)
 - ii. Trial 2 (hypothetical): Therapist presents receptive trial, distracts child for a second, then presents expressive trial
10. **Within Stimulus: Voice Inflection** - the therapist changes how their voice sounds to "highlight" a specific part of a word or to signal that a response is expected. It makes the correct answer stand out or gives the learner a "clue" through the tone of your voice.
 - a. SD: Present object & ask “What color is it?”
 - i. Trial 1: The therapist says, “What color is it?”
 - ii. Trial 2 (hypothetical): The therapist says, “What color is it?”
 - iii. Trial 3 (hypothetical): The therapist says “What color is it?”
 - iv. Trial 4 (hypothetical): The child responds correctly without a voice inflection prompt

Note:

When to prompt:

- ★ When teaching a new target: **0% - 79% correct - Errorless Learning**
 - Prevent errors by using **Most-to-Least (MTL)**
 - When teaching a new skill and acquisition item is 0%-79% correct, a most-to-least prompting hierarchy should be implemented such that prompts occur with the presentation of every SD until the prompts are faded
 - Involves fading from one type of prompt to another by going from the most assistive prompt to the least assistive prompt across trials
 - Example: SD: “Give me book”
 - Full physical prompt (“Give me book” + hand-over-hand)
 - Partial physical prompt (“Give me book” + nudge hand)
 - Gestural prompt (“Give me book” + point to book)
 - NO prompt: SD alone (“Give me book” without a prompt)
 - What if the error occurs during the fading process?
 - Provide an informational “no” (even if the trial was a prompted trial and the child responded incorrectly to the prompt)
- ★ When the target reaches **or above 80% correct - Error Correction**
 - **No-No-Prompt-Repeat**
 - Step 1: Upon error, give an informational “no”
 - Step 2: Repeat SD...If an error occurs again, give an informational “no”
 - Step 3: Repeat SD, and provide the last effective prompt to ensure a correct R
 - Step 4: Repeat SD without a prompt
 - **Least to the most intrusive prompt (LTM)**
 - Step 1: Upon error, give an informational “no”
 - Step 2: Repeat SD and provide the least intrusive prompt...if an error occurs again, provide an informational “no”
 - Step 3: Repeat SD and provide the last effective prompt to ensure a correct R
 - Step 4: Repeat SD without a prompt

Note:

Prompt Fading

How to fade prompts: Prompt fading is the process of systematically removing help so the learner becomes independent.

➤ Set Your Goal (Criteria)

- Do not move to a less intrusive prompt until the learner is successful at the current level.
- *Example:* If the goal is "3 correct responses in a row," stay at the **Partial Physical** level until they hit that mark before moving to a **Gestural** prompt.

➤ **Fade the "Strength" First (Magnitude)**

- Before switching to a completely different prompt type, try making your current prompt weaker.
- *Example:* If you are using a **Gestural** prompt (pointing), start by pointing directly at the object. Then, fade the magnitude by only pointing halfway, then by just glancing at the object.

➤ Switch the Prompt Type

- Once the learner is successful with a "weak" version of a prompt, move to the next level in the hierarchy (e.g., move from Physical to Gestural).

Handling Errors During Fading: If you try to fade a prompt and the learner makes a mistake, follow these steps:

- **Give an Informational "No":** In a neutral, calm voice, say "No," "Not that one," or "Try again." This tells the learner their response was incorrect without being punishing.
- **Backstep:** Immediately represent the instruction (SD) using the **last effective prompt** (the one that worked a moment ago).
- **Try Again:** Once they are successful with the help, begin to fade the prompt again very slowly over the next few turns.

Discrimination training *PLUS* “Prompting” vs. “Fading.”

- Use Most-To-Least prompting when....

- You've faded your prompts out during MT, you may need to reintroduce the last effective prompt when you get to RR
- If the child's supervisor has indicated that either ET or GRR is to be used between MT and RR, you may need to reintroduce the last effective prompt when you begin these variations

- Use an error correction procedure when....

- Error correction procedures should never be used until the acquisition item is 80% correct in RR. At this point, no longer use MTL prompting
- Instead, use the error correction procedure indicated by the child's supervisor

Note:

Stimulus Control vs. Stimulus Transfer

Stimulus Control: The term stimulus control is used to label the relationship between a stimulus and a response when the stimulus (formally labeled discriminated stimulus, SD) sets the occasion on which the response will likely have a consequence (Mayer et al., 2019)

Stimulus Transfer: The process of transferring control of a behavior from a prompt to the natural SD (instruction or cue).

- **Why It Matters:**

- Promotes independent responding
- Prevents prompt dependency
- Ensures the behavior happens in response to real-world cues

- **How it happens:**

- Through prompt fading (e.g., most-to-least, least-to-most)
- Using procedures like time delay, stimulus fading, or differential reinforcement
- Always paired with reinforcement for independent responses

Note:

Stimulus Discrimination VS Stimulus Generalization

Stimulus Discrimination: When SD signals a rich schedule of reinforcement and the S-delta signals zero change of reinforcement (i.e., extinction), the organism will eventually respond only in the presence of the SD and emit no response in the S-delta condition

- Example: When you can tell the difference between colors: Blue vs Red

Stimulus Generalization: The extent to which stimuli other than the SD acquire stimulus control over the behavior. This stimulus shares some similar physical properties with the SD.

- Example: You can know that all different shades of red are red

Note:

The Discrete Trial Training (DTT)

Definition: An instructional technique used in ABA that has a clear beginning and ending (hence, the term discrete)

- It is made up of the 3-term contingency: The antecedent (SD and/or EO), a single response, & a consequence. Example:
 - **Antecedent (A):** Teacher says, “Touch car.”
 - **Response (B):** Jacob touches the car
 - **Consequence (C):** Give Jacob candy

Discriminative stimulus (SD)

Definition: The stimulus is called a discriminative stimulus (SD) because it signals that a correct response to the stimulus will result in reinforcement

Stimulus / Stimuli: Anything that one sees, hears, smells, tastes, or feels

Example:

- Visual stimulus/ SD: A toy cow, picture of a cow, real cow, etc.
- Verbal stimulus/ SD: “What does a cow say?” or “Tell me something that’s red?”
- Visual and Verbal Combined SD:
 - Holding up a toy cow (visual) & asking “What is it?”
 - Pointing to a picture of a boy sleeping (visual) & asking “What is he doing?” (verbal)

Guidelines for giving SD:

- **Have the child’s attention before you present the SD**
 - If you fail to gain a child’s attention, he is unlikely to respond, & you are setting him up for failure
- **Avoid presenting the child’s name with SD**
 - The name may become an extraneous variable that the child “ignores,” OR
 - The child may start to respond only to SDs that are preceded by his or her name
- **Avoid repeating SDs without giving consequences**
 - The child will not respond consistently.
 - We want the child to learn that every time we present an SD, a response is expected
- **Change the target SD only when the supervisor instructs you to do so**
 - Will slow a child’s progress if every therapist works on their own target in a program
- **Initially, SDs should be clear and concise**
 - Imagine you are learning a new language & you want to learn to ask for an apple. Which way will you learn more quickly?
 - ☐ Tx: “Look at this, we call this an apple. Repeat after me -apple-”
 - ☒ Tx: “Apple”
- **The child should respond only after the entire SD is presented**
 - If the child responds before the SD is completed, he or she is not responding to the SD. Therefore, the child is not learning what it is we are intending to teach

Note:

The Inter-Trial Interval (definition)

Definition: The time between the end of one trial and the start of the next

Guidelines:

- Should be long enough to keep each trial distinct but not so long that the child loses attention
- Less than 2 seconds

The Establishing Operation (EO)

Definition: A condition or physical state that alters the effectiveness of a reinforcer. This make the reinforcer either more or less reinforcing

Example:

- **Deprivation:** Food (hungry state), Drink (thirsty state), and Rest (state of fatigue)
- **Satiation:** Feeling Full, recent access to a reinforcer

Note:

The Response (R)

Definition: The specific instance of a particular behavior

Three categories of responses:

- The correct response (C)
- The incorrect response (I)
- No response (NR)

Guidelines for Responses:

- Therapists must be consistent about what is considered a correct response
- Be sure extraneous behavior is absent
- Limit the time between SD and R to 3 seconds
- When the child doesn't respond, consider the NR to be incorrect

Note:

The Consequence (C)

Definition: An event that immediately follows the child's response

Two possible consequences that follow a response:

- Reinforcement
- Error Correction Procedure (*Refer to: "when to prompt's section"*)

The Reinforcer (Sr)

Definition: Immediate presentation of a desired stimulus or removal of an aversive stimulus that increases the future frequency of a response

Categories of Reinforcers:

1. **Primary / Unlearned (Unconditioned Reinforcers):** Food and Drinks
2. **Secondary / Learned (Conditioned Reinforcers)**
 - ★ Praise
 - ★ Rough-housing
 - ★ Toys
 - ★ Tokens
 - ★ A break

Effectiveness of Reinforcers: EOs affect the effectiveness of reinforcers:

- Deprivation of a reinforcer **INCREASES** its effectiveness.
- Satiation of a reinforcer **DECREASES** its effectiveness.

Guidelines for Delivering Reinforcers

- **Limit free access to reinforcers, and instead deliver reinforcers contingent on a target response**
 - If the child has free access to preferred items all day, those items will lose some of their effectiveness as reinforcers
- **Conduct preference assessments frequently (as much as possible)!**
 - Cannot teach without effective reinforcers
 - EO's change over time, thus preferences change during your session
 - *See Appendix 3: Stimulus Preference assessments*
- **Deliver Sr within ½ second of R**
 - Reinforcement is only effective if presented immediately following the R
- **Sr should be easy to give & remove. Why?**
 - Need to be able to present it within ½ s of R
 - If Sr is a food & it takes the child a long time to eat (e.g., gummy candy), we're wasting a lot of valuable teaching time
 - If Sr is a toy & child engages in tantrum when it's taken, teaching time is lost managing the tantrum
- **Establish conditioned (learned) reinforcers**
 - Often when our kids start therapy, social praise & smiles are not preferred
 - Pair so social interaction comes to function as a Sr over time
- **Provide frequent reinforcement throughout the session for appropriate behavior**
 - Every minute of a session is an opportunity to reinforce (and thereby increase) appropriate behavior such as sitting nicely, etc.

How often do I provide the Sr?: Two reinforcement schedules you will use:

1. Continuous Reinforcement
 - Used for teaching new behaviors "on acquisition"
 - Reinforcement occurs following every correct response
2. Intermittent Reinforcement
 - Used to maintain behaviors previously learned or "mastered" (AKA "on maintenance")
 - Reinforcement occurs after every 2, 3, 4, 5, etc. correct responses or after an average number of responses

Ideas/strategies for giving effective vocal praise:

- ★ **Behavior-Specific Praise (Labeling):** Instead of general praise, name the behavior you are reinforcing. This helps the learner connect the reward to the action.
 - Instead of: "Good job!"
 - Say: "Great job sitting with a quiet body!" or "I love how you shared your car with me!"
- ★ **Vary Your Pitch and Energy (The "Big Deal" Factor):** Children with ASD often respond better to "exaggerated" social cues.
 - Low Magnitude: For easy or mastered tasks, use a calm, pleasant voice ("Nice work").
 - High Magnitude: For new, difficult skills (like using the toilet or saying a first word), use high energy, a wide pitch range, and high volume ("WOW! You did it!! That was amazing!").
- ★ **Pair with "Non-Contingent" Conversation:** To make praise feel more natural and less like a "transaction," follow the praise with a comment about the activity.
 - Example: "Awesome drawing! You used so much blue—that's my favorite color."
- ★ **The "Premack" Style Praise:** Combine the praise with the promise of what is coming next to keep motivation high.
 - Example: "You finished your puzzle so fast! Now it's time for bubbles!"

Note:

Natural Environment Training (NET)

Natural Environment Training (NET): Teaching procedures within Applied Behavior Analysis (ABA) that can be used to teach learners in their natural environments.

- Therapists capitalize on the learner's motivation in a natural situation. Teaching is incorporated into play activities using familiar toys, games, and materials to maximize the learner's motivation to continue the activity.

Key Components:

- **Motivation (MO):** We follow the child's lead. If they want the bubbles, we teach using bubbles.
- **Reduce Problem Behavior:** Because the child is doing what they like, they are less likely to engage in escape-based behaviors.
- **Functional Learning:** Skills are learned through real-life experience and "doing," rather than rote memorization.
- **Natural Generalization:** Skills learned in play are more likely to happen again in the "real world" without extra training.

Reinforcement in NET is Dynamic and Functional:

- **The "Natural" Connection:** If the child asks for a "Swing," the reinforcement is getting a push on the swing.
- **Dynamic Choice:** Because a child's interests change quickly during play, the therapist must be ready to switch reinforcers instantly as the child's motivation shifts.
- **Building Social Value:** In NET, the therapist becomes a "giver of fun," which helps the child value social interaction more than just the toy.

Note:

Verbal Operants

Definition: A behavior reinforced through the mediation of other persons. It includes any spoken or non-spoken form of communication that helps people to get what they want and avoid what they do not want (Mayer et al., 2019)

Four types of verbal operants:

1. **Echoic:** Echoic involves the speaker repeating the trainer's sounds, words, or phrases.
2. **Mand:** A speaker to communicate their needs and wants to a listener.
3. **Tact:** Acts refer to a speaker's skill in verbally identifying features of their physical environment.
4. **Intraverbal:** Intraverbal repertoires allow us to verbally respond to verbal stimuli we encounter or produce ourselves.

Note:

Different Teaching Strategies

Imitation: Matching the behavior of a model, or engaging in a behavior similar to that observed

Modeling: An antecedent stimulus with topographical similarity to the behavior the analyst wants to be imitated

Demand Fading: Gradually increasing task demands from easy(number/duration/difficulty) to challenging until an individual can meet age-appropriate or target-level expectations without problem behavior.

★ Demand: Any task or or assignment or activity that the learner is asked to do or complete

Pairing: Associates yourselves with preferred activities and reinforcement so your presence becomes reinforcing. During pairing, you place few or no demands, follow the client's lead, and frequently deliver preferred items or activities. This helps build rapport, increase cooperation, and establish instructional control for future teaching.

Note:

Shaping and Chaining

Shaping

Definition: Reinforcing successive approximations of a target behavior while extinguishing previous approximations

When to use: When the target behavior doesn't occur at all & there are no prompts we can use to get the behavior to occur

How to Shape Behaviors:

- **Step 1:** Select the target behavior (e.g., "car")
- **Step 2:** Establish the initial behavior (e.g., "c")
- **Step 3:** Reinforce initial behavior until it is under SD control (e.g., Jimmy says "c" consistently every time the therapist presents the SD "say car")
- **Step 4:** When the child learns a better approximation (e.g., "ca"), no longer reinforce the previous approximation
- **Step 5:** Continue until the child learns the target behavior

Note:

Chaining:

Definition: A sequence of SDs and Rs where each R in the sequence (except the last) produces the next SD and the last R is typically followed by a reinforcer

Task Analysis (definition): Break down a complex behavior into a sequence of components

Types of Chaining (3 types)

1. Forward Chaining

- In a sequence of SDs and Rs, the first R is taught 1st
- Then, the 2nd R is taught
- Then, the two are performed independently in order
- Then, the 3rd R is taught & performed in sequence with the 1st & 2nd R
- And so on until all the behaviors in the chain are performed independently

2. Backward Chaining

- In a sequence of SDs and Rs the last R is taught 1st, then the 2nd to the last, then the two are performed independently in order
- Then the 3rd to the last R is taught & performed in sequence with the last two Rs
- And so on until all the behaviors in the chain are performed independently

3. Total Task Presentation: The simultaneous training of all links in a behavioral chain

Note:

Maintenance and Generalization

Maintenance

Definition: Continued performance of a response after it was first learned

How often do I reinforce:

- **New Skills:**
 - Continuous Reinforcement: Early on, reinforcement occurs after every attempted or correct trial.
 - Intermittent reinforcement: As children progress, we reinforce less frequently, after every 2, 3, 4, 5, etc. trials.
- **Mastered Skills:**
 - To maintain a skill that is already mastered (and therefore easier for the child) we use an intermittent schedule of reinforcement.

Generalization

Definition: The occurrence of the relevant behavior under different, untrained conditions

Types of Generalization:

1. Across Stimuli: Visual, Verbal
2. Across Setting: Home, School
3. Across People: Therapist, Parent, Teacher

Behavior Management

Behavior Intervention Plan (BIP)

Components of BIP:

- **Operational Definition:** A clear definition of the behavior, example and non-example of target behavior
- **Identify Function:** Identify cause of behavior
- **Antecedent modification:** Strategies aim to prevent behavior by altering the environment, modifying demands, or teaching expectations.
- **Replacement behavior:** The behavior we want the learner to do instead of the problem behavior - one that serves the same function in a more appropriate way.
- **Data collection methods:** How to take data and/or measure the behavior

Note:

Function of Behavior

Definition: Cause of behavior (why the behavior occurs)

Four Controlling Variables:

1. Social Positive Reinforcement

➤ **Attention** Social Positive Reinforcement:

- Child gets good stuff from someone
- Someone giving attention for problem behavior
- Vocal attention: reprimands, statements of concern, lecture or discussion
- Facial expressions: eye contact, glaring, looks of concern or surprise
- Touching: pats, rubs, spansks

➤ **Access to tangible item/activity** Social Positive Reinforcement:

- Child gets good stuff from someone
- Someone giving access to a tangible item/activity for problem behavior
- Toys
- Food
- Activities: getting to go outside, watch TV., spend the night at a friend's house

2. Social Negative Reinforcement

➤ **Escape** Social Negative Reinforcement:

- Someone lets child escape bad stuff
- Someone allowing escape from an aversive event or task contingent upon problem behavior
- Discontinuing a demand or task
- Leaving a setting: child doesn't want to be at school, so child tantrums, and teacher sends child home

3. Automatic Positive Reinforcement

- Behavior itself feels good and is reinforcing
 - Sensory input: child may engage in stereotypy to receive sensory input (e.g., hand flapping, delayed echolalia, twirling)

4. Automatic Negative Reinforcement

- Behavior itself let's child escape something that feels bad
 - Pain relief: child could engage in head banging to relieve (escape) a headache or biting to relieve (escape) a toothache

How to Identify Function?: Functional Behavior Assessment (FBA) - 3 types

1. Indirect Observation: Interview, checklist, questionnaire
2. Direct Observation: ABC data (*Frequently used at CARE*)
3. Functional (Experimental) Analysis (FA): A systematic method to control antecedent and consequence to identify which consequence maintains the behavior.

Note:

Behavior Management Techniques

Antecedent Modification

Definition: An environmental condition or stimulus change existing or occurring prior to a behavior of interest

- ★ An antecedent intervention is implemented prior to and independent of the target behavior's occurrence

Why do we need Antecedent Modification?

- ★ To prevent the behavior of interest (mostly undesired behavior) to occur
- ★ Allow the therapist to teach an expected behavior and the learner to also learn an expected behavior

Commonly Used Antecedent Strategies:

- **Noncontingent Reinforcement (NCR):** an antecedent intervention that may be used to reduce problem behavior
 - Access to reinforcers are delivered on a time-based schedule, independent of the problem behavior
 - Basically, we are giving the child what they want for free
 - If we are providing the reinforcer for free, it should not be necessary for the child to engage in problem behavior to gain access to your attention or a toy that they want
 - Example: Attention: Mom gives praise or checks in every 5-10 minutes, even without a request
 - If we are providing stimulating items to engage with for free, NCR may also make it less likely for some children to engage in behaviors maintained by automatic reinforcement
 - Example: Every 20 minutes, the teacher allows the student to visit the sensory corner for a quick sensory break. Fidget toy can also be provided
- **Teaching Replacement Behaviors:** Teach functionally equivalent replacement behavior with a lower-effort way of getting what they want. This leads to a reduction in undesired behaviors.
- **Premack Principle:** A principle that states that making the opportunity to engage in a high-probability behavior contingent on the occurrence of a low-frequency behavior will function as reinforcement for the low-frequency behavior. Sometimes called "Grandma's Law."
 - Example: The teacher tells the student, "First, you finish this worksheet, then you'll have 10 minutes to play with the class iPad."
- **Demand Fading:** Start with shorter tasks or fewer demands and gradually increase them as the student's tolerance improves.
 - Example: Instead of giving a long worksheet, give the student a shorter version and gradually add more as they can handle it without trying to escape.
- **Task Modification:** Embed preferred activities or interests into less-preferred tasks to make them more engaging. "Changing some aspect of task"
 - Example: If the student enjoys drawing but dislikes math, the teacher could allow them to solve math problems by drawing out the answers or creating a math-related drawing as part of the task.
- **Choices:** Offering two or more acceptable options to increase a learner's sense of control, promote engagement, and support cooperation — while still meeting the intended goal or task.
 - Example: Do you want to work at the table or on the rug?"
- **Behavior Momentum:** An antecedent intervention in which two to five easy tasks with a known history of learner compliance (the high-p requests) are presented in quick succession immediately before requesting the target task, the low-p request.

- Example:
 - Teacher: What is in your hand? Student: Pencil
 - Teacher: You are right!
 - Teacher: Touch your nose. Student: Touches your nose
 - Teacher: Great work—awesome listening
 - Teacher: Write your name. Student: Start writing his name
 - Teacher: That is fantastic—great job!

Common Environment and Visual Supports:

- **Functional Communication Training (FCT):** An antecedent intervention in which an appropriate communicative behavior is taught as a replacement behavior for problem behavior usually evoked by an establishing operation (EO); involves differential reinforcement of alternative behavior (DRA).
 - Example:
 - Escape: A student typically screams when work is too hard. With FCT, they're taught to say "Can I have a break?" instead
 - Attention: A child who hits to get attention is taught to say "Look at me!" or raise their hand to gain appropriate attention.
- **Alternative Functional Communication:** The form of communication extends beyond just speaking; there are various alternative ways for our clients to "express" themselves
 - Picture exchange communication system (PECS)
 - Augmentative and Alternative Communication (AAC)
- **Social Story:** A Social Story is a short, simple story that describes a situation, skill, or concept using clear language and relevant social cues, written from the learner's point of view.
- **Visual Support:** Uses images, symbols, or visual aids to help students understand instructions, routines, or concepts. It's especially helpful for students who struggle with verbal instructions, as it reinforces learning and provides clarity.
- **Transitional Warning:** A proactive strategy that alerts students to an upcoming change in activity. These cues help learners mentally prepare, reduce anxiety, and support smoother transitions through tools like timers, countdowns, or verbal prompts.

Note:

Replacement Behavior:

Definition: The behavior we want the learner to do instead of the problem behavior — one that serves the same function in a more appropriate way.

Why do we need this?

- Teaches a new, acceptable behavior to meet the learner's needs
- Must be easy to learn and required lower effort compare to the behavior of concern
- Most replacement behaviors involve some form of communication
- More likely to last if they consistently help the learner achieve their goals.

Key Consideration:

- Functional Equivalent to target behavior
- Being reinforced consistently, especially in the beginning

Note:

Consequence Modification:

- **Differential Reinforcement:** This is the process of reinforcing only specific behaviors while withholding reinforcement for others.
 - **Differential Reinforcement of Other Behavior (DRO):** Reinforcement is given if the problem behavior does not occur for a specific amount of time. You are reinforcing "anything else."
 - *Example:* A child screams for attention. The therapist sets a timer for 2 minutes. If the child does not scream for the full 2 minutes, they get a sticker.
 - **Differential Reinforcement of Alternative Behavior (DRA):** Reinforcing a specific, appropriate behavior that serves the same purpose as the problem behavior.
 - *Example:* A child hits to get a toy. The therapist stops reinforcing the hitting and only gives the toy when the child uses a "Mand" (e.g., says "Toy please" or taps the therapist's shoulder).
 - **Differential Reinforcement of Incompatible Behavior (DRI):** A type of DRA where the new behavior is physically impossible to do at the same time as the problem behavior.
 - *Example:* A child engages in "hand-mouthing" (putting fingers in mouth). The therapist reinforces the child for keeping their hands in their pockets. They cannot have their hands in their pockets and their mouth at the same time.

- **Token Economy:** A system where a learner earns "tokens" (stars, points, stickers) for specific behaviors. These tokens are later exchanged for a "back-up reinforcer" (a toy, a break, or a snack).
 - **Practice Tip:** Always "pair" the delivery of a token with **Vocal Praise** so the child understands exactly why they earned it.
 - *Example:* Present "A Token board" where the child earns 5 stars for working on math to earn a 5-minute iPad break.
- **Extinction:** the process of no longer providing the reinforcer for a behavior that was previously reinforced. This leads to a decrease in the behavior over time. *Example.* a child usually gets a "hug" (attention) for crying, during extinction, the therapist remains neutral and provides no eye contact or physical touch when the crying occurs.
 - **Extinction burst:** When you first start extinction, the behavior will often get worse before it gets better.
 - **Rule of Practice:** Do not give up during the burst! If you reinforce the behavior during the burst, you are accidentally teaching the child that they just need to "scream louder" to get what they want.
 - **Applying Extinction with Compassion and Dignity:** Extinction can be difficult for both the learner and the therapist. To maintain our ethical commitment to the client's dignity, follow these compassionate guidelines:
 - **It is "Planned Ignoring" of the Behavior, Not the Child:** Even when a behavior is on extinction, the child should never feel abandoned. You are ignoring the *screaming*, not the *human being*.
 - **Maintain Safety and Proximity:** Stay close enough to ensure the child is safe. Moving away slightly is okay to reduce attention, but leaving a child alone in distress violates their dignity.
 - **Monitor Emotional Well-being:** If a child is having an extinction burst and becomes highly distressed, remain a "calm presence." Your neutral body language should signal: *"I am here, you are safe, but this behavior will not get you what you want."*
 - **Validate, Don't Negotiate:** You can validate a child's feelings without giving in to the behavior. *Example:* "I can see you are frustrated that it's time to work, but we still need to finish this puzzle." (Then follow through with Escape Extinction).
 - **Important Practice Rules for Staff:**
 - **Safety First:** Never use extinction for dangerous behaviors (self-injury or aggression) without a supervisor-approved safety plan. If a child's safety is at risk, safety protocols override extinction.
 - **Consistency is Kindness:** Being inconsistent (giving in sometimes) is actually unkind, as it confuses the child and makes their struggle last longer. Absolute consistency helps the child learn the new "rule" faster.

Note:

Punishment

Definition: any consequence that decreases the future frequency of a behavior. Because punishment involves adding something aversive or taking away something preferred, it must be handled with the highest level of ethical care.

Punishment Procedures

- **Nonexclusionary Time-Out:** Immediate, response contingent removal of access to a reinforcer resulting in a decreased frequency of that response.
 - **Points to Remember**
 - A reinforcing activity must be in effect prior to time out, otherwise time out will be an opportunity for escape or avoidance
 - Time-out should never involve closing a child into a room as this would be considered exclusionary and an aversive
 - Time out should be immediate
 - No attention should be given during time out
 - Duration of time out depends on the age of your child (1 min per 1 year of age)
 - When time is up, if your child is engaging in any inappropriate behavior, wait for him or her to behave appropriately for at least 15 s before removing from time out
- **Response Cost:** Immediate, response contingent removal of a portion of a tangible reinforcer resulting in a decreased frequency of that response
- **Over-correction:** The learner not only fixes what they did (restitution) but also repeats appropriate behavior in an exaggerated way.
 - It adds effort after the problem behavior to reduce future occurrences.
- **Positive Practice:** The learner repeats the correct behavior many times after doing it incorrectly.
 - It adds extra effort to reduce the likelihood of the problem behavior happening again.

Caution When Using Punishment!

- Use ethically and under supervision
- Never use punishment as a first-line strategy
- Continue to reinforce appropriate behavior
- The goal is not to control, but to support meaningful behavior change
- Watch for Side Effects: Improper use of punishment can lead to:
 - Escape/avoidance of adults
 - Emotional responses
 - Increased aggression
- Monitor data and ensure it's effective and humane

Note:

Data Collection

Why do we need it?

- To make the decision-driven decision
 - Show if the treatment is working
 - Guides daily decision
 - Making decision based on the observable and measureable behavior, not assuming
- It is a part of assessment: Preference assessment, Functional Behavior Assessment (FBA)
 - ABC data recording:
 - Structured ABC data recording: Antecedent and Consequence list are predetermined by supervisor
 - Unstructured ABC (descriptive) data recording: the therapist can describe what antecedent and consequence themselves

Type of Data Collection:

Continuous: Observing and recording all instances of behavior (e.g., entire session)

- **Frequency:** To measure a behavior is simply to count how many times it happens
 - **Rate:** Dividing the frequency of the observation by the unit of time (e.g., minute, hour) that the observation lasted
 - **Percentage:** How often a behavior occurs relative to opportunities or total trials.
- **Duration:** Measure how long the behavior lasts
- **Latency:** How long it takes the behavior to occur after the instruction is presented
- **Interresponse Time (IRT):** The amount of time between two consecutive occurrences of the same behavior.

Discontinuous: Observing behavior at specific intervals or moments

- **Whole Interval:** Observation time is divided into equal intervals (e.g., 10 seconds, 1 minute). You only mark the interval as “behavior occurred” if the behavior lasted for the full interval
- **Partial Interval:** Divides observation time into equal intervals (e.g., 10 seconds, 1 minute). Behavior does not have to occur for the full interval — just once within it
- **Momentary Time Sampling:** Time is split into equal intervals (e.g., 1 minute, 30 seconds). Behavior is observed only at the interval's end. Behavior must occur at that moment to record it.
 - Useful for group or classroom behavior assessment

Note:

Data Analysis:

Graph:

- Data are recorded onto datasheets in the logbook and then plotted onto data analysis graphs, both for skills in acquisition and for challenging behavior
- Data must be collected and graphed on a daily basis, so they can be visually inspected and treatment decisions can be made on the basis of the data
- Data are tracked using whatever measurement the clinical supervisor puts in place for the lesson
- Once the data are recorded, therapists convert the data into a form of measurement as indicated by the clinical supervisor and then plot the data onto the data analysis graphs
- Depending on what the data are tracking, this could include plotting a percentage correct for a block of trials, frequency of a behavior, rate per minute/hour for a behavior, and so on.

Key Elements:

- **Vertical Axis (Y-axis):** Shows the behavior being measured (e.g., number of responses, duration)
- **Horizontal Axis (X-axis):** Represents time, usually in sessions, days, or weeks
- **Marks:** Small marks on each axis to separate units of measurement
- **Axis Labels:** Clearly describe what each axis represents (e.g., "Frequency" or "Sessions")
- **Data Points:** Dots plotted to show the behavior value for each session
- **Data Paths (Lines):** Connect data points to show behavior trends over time
- **Phase Lines:** Vertical lines that show a change in condition (e.g., baseline → intervention)
- **Phase Labels:** Text above each phase (e.g., "Baseline", "Intervention", "Follow-Up")
- **Legend:** Explains symbols or multiple data paths when more than one is shown

Data Aspects:

- **Level:** How high or low the data are. A rapid change in the level of the data in the direction that the treatment intends is the strongest indication that the intervention is working
- **Trend:** The overall direction of the data path — whether it's increasing, decreasing, or stable over time. Many effective interventions do not produce immediate change in level, but they do produce a positive trend that is detectable within a few sessions of intervention
- **Variability:** The degree of fluctuation or spread of the data points. It reflects how consistent the behavior is across sessions.

Interobserver Agreement (IOA):

- The degree to which two or more observers report the same observed values for the same behavior.
- It ensures that behavior is being measured consistently and objectively — no matter who collects the data.
- Example:
 - Two observers watch the same session.
 - If Observer A records 5 instances of hitting, and Observer B records 5 too → IOA = 100%.

Note:

Appendix

Appendix 1: Child Abuse & Mandated Reporting

Child Abuse

What Is Child Abuse and Neglect?

Recognizing the Signs and Symptoms. The first step in helping children who have been abused or neglected is learning to recognize the signs of maltreatment. The presence of a single sign does not necessarily mean that child maltreatment is occurring in a family, but a closer look at the situation may be warranted when these signs appear repeatedly or in combination. This factsheet is intended to help you better understand the Federal definition of child abuse and neglect; learn about the different types of abuse and neglect, including human trafficking; and recognize their signs and symptoms. It also includes additional resources with information on how to effectively identify and report maltreatment and refer children who have been maltreated.

The Federal Child Abuse Prevention and Treatment Act (CAPTA), as amended and reauthorized by the CAPTA Reauthorization Act of 2010, defines child abuse and neglect as, at a minimum, “any recent act or failure to act on the part of a parent or caretaker which results in death, serious physical or emotional harm, sexual abuse or exploitation (including sexual abuse as determined under section 111), or an act or failure to act which presents an imminent risk of serious harm” (42 U.S.C. 5101 note, § 3).

Major Types of Child Abuse and Neglect:

Most recognized in these *four major types of maltreatment*: physical abuse, neglect, sexual abuse, and emotional abuse. Additionally, many may also identify *abandonment, parental substance use, and human trafficking as abuse or neglect*. While some of these types of maltreatment may be found separately, they can occur in combination. This section provides brief definitions for each of these types.

1. Physical abuse

- A nonaccidental physical injury to a child caused by a parent, caregiver, or other person responsible for a child can include punching, beating, kicking, biting, shaking, throwing, stabbing, choking, hitting (with a hand, stick, strap, or other object), burning, or otherwise causing physical harm.
- Physical discipline, such as spanking or paddling, is not considered abuse as long as it is reasonable and causes no bodily injury to the child.
- Injuries from physical abuse could range from minor bruises to severe fractures or death.

2. Sexual abuse

- Sexual abuse includes activities by a parent or other caregiver, such as fondling a child’s genitals, penetration, incest, rape, sodomy, indecent exposure, and exploitation through prostitution or the production of pornographic materials.
- Sexual abuse is defined by CAPTA as “the employment, use, persuasion, inducement, enticement, or coercion of any child to engage in, or assist any other person to engage in, any sexually explicit conduct or simulation of such conduct for the purpose of producing a visual depiction of such conduct; or the rape, and in cases of caretaker or interfamilial relationships, statutory rape, molestation, prostitution, or other form of sexual exploitation of children, or incest with children

3. Neglect

- Neglect is the failure of a parent or other caregiver to provide for a child’s basic needs. Neglect generally includes the following categories:

- Physical (e.g., failure to provide necessary food or shelter, lack of appropriate supervision)
 - Medical (e.g., failure to provide necessary medical or mental health treatment, withholding medically indicated treatment from children with life-threatening conditions)
 - Educational (e.g., failure to educate a child or attend to special education needs)
 - Emotional (e.g., inattention to a child's emotional needs, failure to provide psychological care, permitting a child to use alcohol or other drugs)
 - Sometimes cultural values, the standards of care in the community, and poverty may contribute to what is perceived as maltreatment, indicating the family may need information or assistance.
 - It is important to note that living in poverty is not considered child abuse or neglect. However, a family's failure to use available information and resources to care for their child may put the child's health or safety at risk, and child welfare intervention could be required.
- 4. Emotional abuse (or psychological abuse)**
- Emotional abuse (or psychological abuse) is a pattern of behavior that impairs a child's emotional development or sense of self-worth.
 - This may include constant criticism, threats, or rejection as well as withholding love, support, or guidance. Emotional abuse is often difficult to prove, and, therefore, child protective services may not be able to intervene without evidence of harm or mental injury to the child

Abandonment

Abandonment is considered a form of neglect. In general, a child is considered to be abandoned when the parent's identity or whereabouts are unknown, the child has been left alone in circumstances where the child suffers serious harm, the child has been deserted with no regard for his or her health or safety, or the parent has failed to maintain contact with the child or provide reasonable support for a specified period of time.

Parental substance

Parental substance use is included in the definition of child abuse or neglect. Related circumstances that are considered abuse or neglect include the following:

- Exposing a child to harm prenatally due to the mother's use of legal or illegal drugs or other substances
- Manufacturing methamphetamine in the presence of a child
- Selling, distributing, or giving illegal drugs or alcohol to a child
- Using a controlled substance that impairs the caregiver's ability to care for the child adequately

Human trafficking

Human trafficking is considered a form of modern slavery and includes both sex trafficking and labor trafficking. Sex trafficking is recruiting, harboring, transporting, providing, or obtaining someone for a commercial sex act, such as prostitution, pornography, or stripping. Labor trafficking is forced labor, including drug dealing, begging, or working long hours for little pay (Child Welfare Information Gateway, 2018). Although human trafficking includes victims of any sex, age, race/ ethnicity, or socioeconomic status, children involved in child welfare, including children who are in out-of-home care, are especially vulnerable.

Recognizing Signs of Abuse and Neglect and When to Report:

It is important to recognize high-risk situations and the signs and symptoms of maltreatment. If you suspect a child is being harmed, reporting your suspicions may protect him or her and help the family receive assistance. Any concerned person can report suspicions of child abuse or neglect. Reporting your concerns is not making an accusation; rather, it is a request for an investigation and assessment to determine if help is needed.

A child may directly disclose to you that he or she has experienced abuse or neglect. If so, please inform your supervisor and local law enforcement

- ★ Child Protection Foundation(มูลนิธิศูนย์พิทักษ์สิทธิเด็ก): 0-2412-1196.
- ★ Social Assistance Center(ศูนย์ช่วยเหลือสังคม): 1300.

While it's important to know the signs of physical, mental, and emotional abuse and neglect, which are provided later in this factsheet, the following signs of general maltreatment can also help determine whether a child needs help:

Children's signs:

- Shows sudden changes in behavior or school performance
- Has not received help for physical or medical problems brought to the parents' attention
- Has learning problems (or difficulty concentrating) that cannot be attributed to specific physical or psychological causes
- Is always watchful, as though preparing for something bad to happen
- Lacks adult supervision
- Is overly compliant, passive, or withdrawn
- Comes to school or other activities early, stays late, and does not want to go home • Is reluctant to be around a particular person
- Discloses maltreatment

Parents' sign:

- Denies the existence of—or blames the child for—the child's problems in school or at home
- Asks teachers or other caregivers to use harsh physical discipline if the child misbehaves
- Sees the child as entirely bad, worthless, or burdensome
- Demands a level of physical or academic performance the child cannot achieve
- Looks primarily to the child for care, attention, and satisfaction of the parent's emotional needs
- Shows little concern for the child
- Parent and child
- Touch or look at each other rarely
- Consider their relationship entirely negative
- State consistently, they do not like each other

Signs of Physical Abuse: A child who exhibits the following signs may be a victim of physical abuse:

- Has unexplained injuries, such as burns, bites, bruises, broken bones, or black eyes
- Has fading bruises or other noticeable marks after an absence from school
- Seems scared, anxious, depressed, withdrawn, or aggressive
- Seems frightened of his or her parents and protests or cries when it is time to go home
- Shrinks at the approach of adults
- Shows changes in eating and sleeping habits
- Reports injury by a parent or another adult caregiver
- Abuses animals or pets

A parent or other adult who exhibits the following signs:

- Offers conflicting, unconvincing, or no explanation for the child's injury or provides an explanation that is not consistent with the injury
- Shows little concern for the child
- Sees the child as entirely bad, burdensome, or worthless
- Uses harsh physical discipline with the child
- Has a history of abusing animals or pets

Signs of Sexual Abuse: A child who exhibits the following signs may be a victim of sexual abuse:

- Has difficulty walking or sitting
- Experiences bleeding, bruising, or swelling in their private parts
- Suddenly refuses to go to school
- Reports nightmares or bedwetting
- Experiences a sudden change in appetite
- Demonstrates bizarre, sophisticated, or unusual sexual knowledge or behavior
- Becomes pregnant or contracts a sexually transmitted disease, particularly if under age 14
- Runs away
- Reports sexual abuse by a parent or another adult caregiver
- Attaches very quickly to strangers or new adults in their environment

Signs of Emotional Maltreatment: A child who exhibits the following signs may be a victim of emotional maltreatment:

- Shows extremes in behavior, such as being overly compliant or demanding, extremely passive, or aggressive
- Is either inappropriately adult (e.g., parenting other children) or inappropriately infantile (e.g., frequently rocking or head-banging)
- Is delayed in physical or emotional development
- Shows signs of depression or suicidal thoughts
- Reports an inability to develop emotional bonds with others

Appendix 2: Branches of Behavior Analysis - Behavior Analysis is a broad field with four distinct branches

Branches of Behavior Analysis: (Cooper et al., 2019, Chapter 1, P7, 10, P20)

- **Behaviorism:** The theoretical and philosophical foundation underlying Applied Behavior Analysis (ABA) techniques
 - The behaviorist perspective holds that *behavior is learned and can be observed and measured objectively*, rather than internal mental states
 - *The behavior is shaped by environmental influences.* Understanding behavior, therefore, requires direct observation of the relationship between environmental stimuli (S) and behavioral responses (R). This stimulus-response framework emphasizes that changes in the environment can lead to predictable changes in behavior
- **Experimental Analysis of Behavior (EAB):** The philosophy and theory of behaviorism is experimented under controlled conditions called basic research.
 - EAB seeks to identify the basic principles that govern behavior through *systematic experimentation*.
 - EAB emphasizes the *direct manipulation of environmental variables* to examine their effects on behavior.
 - EAB primarily investigates *operant conditioning*, where behavior is shaped and maintained by its consequences
 - Behaviors followed by reinforcement increase
 - Behaviors followed by punishment decrease

	Behavior INCREASE (Reinforcement)	Behavior DECREASE (Punishment)
Stimuli is ADDED (Positive)	Positive Reinforcement Ex: Hungry >> baby cries >> mom feeds >> more likely to cry again	Positive Punishment Ex: Skip class >> Reprimand >> Stop/decrease
Stimuli is REMOVED (Negative)	Negative Reinforcement Ex: Hot water >> Jump out >> Escape from burning water >> more likely to jump out again	Negative Punishment Ex: Skip class >> Take away play time >> Stop/decrease

- **Applied Behavior Analysis:** The researchers' major purpose was to determine whether the basic principles of behavior discovered in the laboratory operated with humans
 - **Characteristic of Applied Behavior Analysis: Baer, Wolf, and Risley (1968)**
 - **Applied:** Socially Significant
 - **Behavior:** Observable and Measurable
 - **Analytic:** A demonstration of a functional relationship between environment and behavior
 - **Technological:** Procedures are described with sufficient detail and clarity
 - **Conceptually systematic:** the procedures should be described in terms of the relevant principle(s)
 - **Effective:** An effective application of behavioral techniques must improve the behavior under investigation to a practical degree
 - **Generality:** A behavior change lasts over time, appears in environments other than the one in which the intervention that initially produced it was implemented, and/or spreads to other behaviors not directly treated by the intervention.
 - **Practice Guided by Behavior Analysis:** professional practice in various fields that is informed and guided by that science (but does not cover all 7 characteristics of ABA)

Appendix 3: Stimulus Preference assessments

Why do we need it?

- To determine what may be reinforcing to the client
- If we use effective reinforcers, the likelihood that the client will engage in that same behavior in the future increases

Types of Stimulus Preference Assessments

- **Ask:**
 - Client: Ask the client to identify preferred item/reinforcer
 - Significant others: Ask the person who are close to the client to identify preferred item/reinforcer
- **Free-operant:**
 - Contrived: Contrived the setting and observing the person approaching, interacting, or engaging with various stimuli
 - Natural: observing the person approaching, interacting, or engaging with various stimuli in client's natural environment
- **Structured Preference Assessment:** Conduct an assessment that provides the individual opportunities to indicate their preferences through physical choice responses
 - **Multiple Stimuli Without Replacement (MSWO):** A preference assessment where the learner chooses from multiple items presented at once, and the selected item is removed from the array in the next trial.
 - **Multiple Stimuli With Replacement (MSW):** A preference assessment where the learner chooses from multiple items, but the selected item is returned to the array on the next trial.
 - **Paired Choice:** A systematic preference assessment where the learner is presented with two items at a time and asked to choose one.